PS-3 run report, host A: bit-sliced group-law kernel at 64, 256, 512 lanes

PDP oracle speedup program, `research/pdp_oracle_speedup_20261011/ps3_bitsliced/raw/` — 2026-10-12, 02:16–02:52 UTC

1. What ran

`ps3_bitsliced/run_ps3.sh` (correctness, programs, utilisation, product micro-benchmark, then the interleaved timing stage), `examples/pdp_speedup`, one release binary for every arm (SHA-256 `c40434d6...a6c1`), runtime dispatch, no global `target-cpu`. Host A: Intel Xeon 2.80 GHz KVM guest, 4 vCPU, AVX-512 F/BW/CD/DQ/VL/VNNI, PCLMULQDQ, kernel `6.18.44-fc-v114`; timed runs on CPU 3 through `tools/isolated_bench.py run --cpus 3` (a shared container, not the repository's receipt service). Arms: `baseline` (`decompose`), `baseline2` (the A/A), `group-scalar` (PS-2), `bs64`, `bs256`, `bs512`. Five rounds requested per rung, every round runs every arm once.

Runs requested and completed. 180 timed rows requested (6 arms x 5 rounds x 6 rungs `l = 8..13`); **169 completed**. The container stopped during `l = 13`, round 4; `l = 13` has three complete rounds. 4 of the 169 are marked contended by the tool (`baseline2-l8-r2`, `group-scalar-l8-r2`, `group-scalar-l9-r2`, `bs64-l9-r2`); they are kept in every figure and table, and `ab_summary_excl_contended.md` drops their rounds. 9 starts were refused by the pre-flight and retried (`raw/refused_starts.log`). The thread-scaling stage did not run on this host.

2. Observations

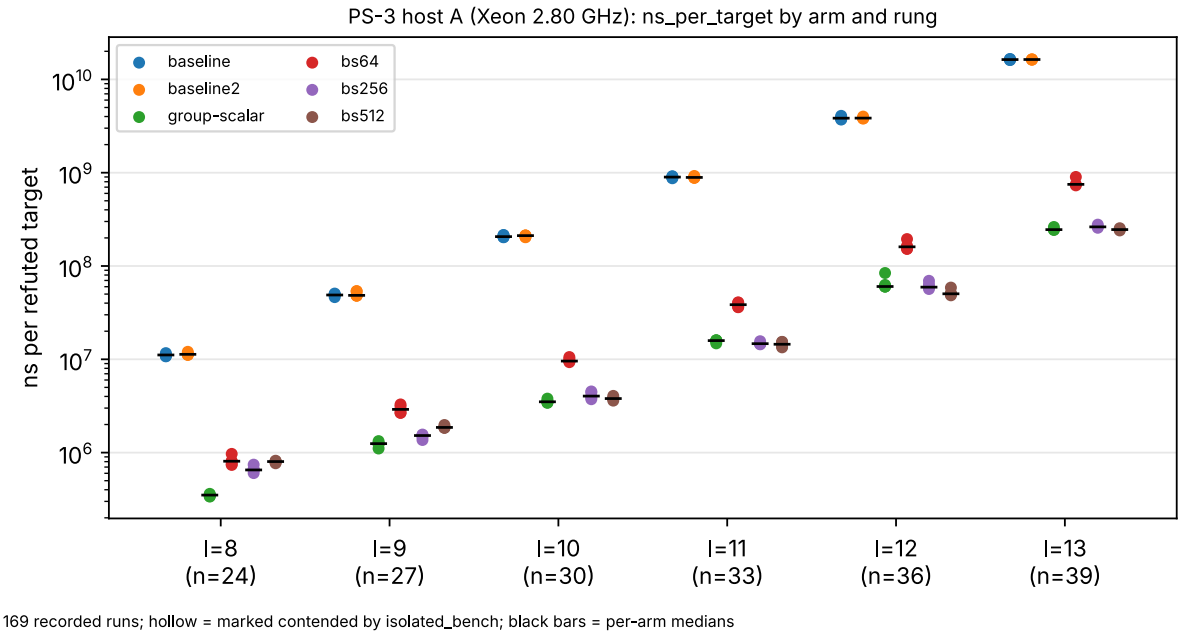


Figure 1: Wall nanoseconds per refuted target by arm and rung (source `raw/times.jsonl`; hollow marks are runs the tool marked contended; black bars are per-arm medians; log scale).

l	baseline ms	group-scalar	bs64	bs256	bs512	rounds
8	11.13	0.351	0.808	0.652	0.802	5
9	48.88	1.248	2.906	1.524	1.862	5
10	206.44	3.504	9.574	4.035	3.796	5
11	897.32	15.863	38.494	14.711	14.498	5
12	3839.87	60.292	160.481	59.347	50.358	5
13	16167.60	245.456	749.956	262.647	245.700	3

Medians over rounds, milliseconds per refuted target. A/A (baseline against baseline) run-to-run spread: 7.95%, 8.72%, 5.42%, 5.01%, 9.30%, 2.06% at $l = 8..13$.

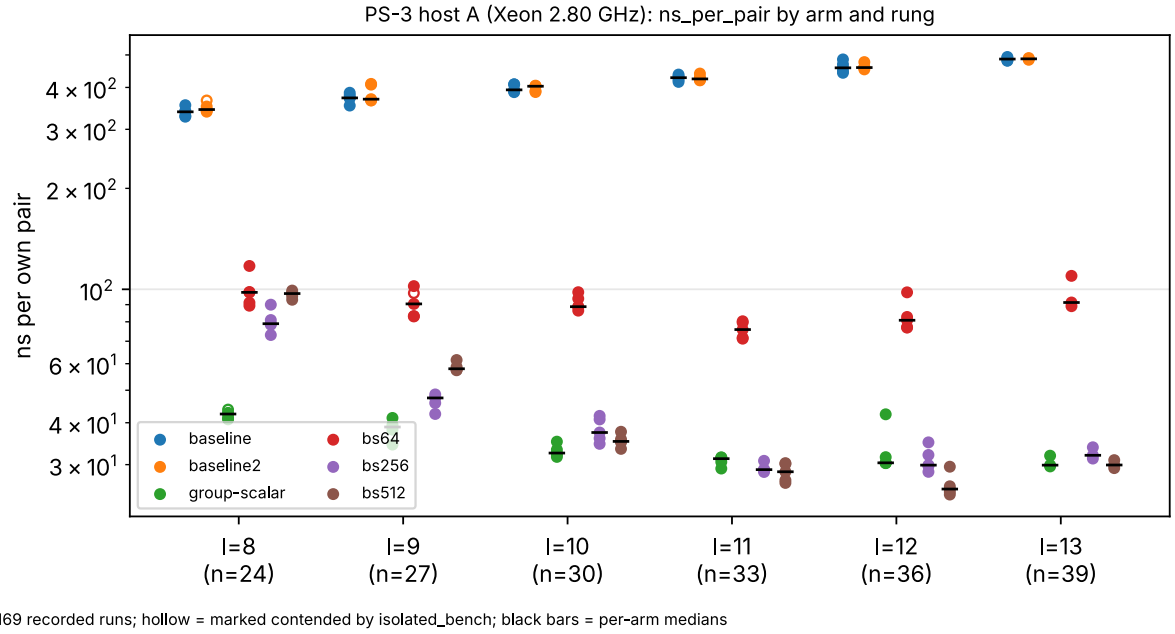


Figure 2: Wall nanoseconds per own pair (an arm's pairs are its own: the baseline visits every abscissa of V, the others the liftable base points).

At $l = 11$ the baseline against each arm, per target (B/A medians, reciprocals derived): group-scalar 0.0175 (57.1x), bs64 0.0420 (23.8x), bs256 0.0165 (60.6x), bs512 0.0160 (62.5x). Against the scalar group kernel, per own pair, $l = 11$: bs256 0.934, bs512 0.935 (min..max 0.841..1.038; the A/A spread of the scalar arm is 7.48%).

Verdict digests equal PS-0′s on every rung for all six vector arm/thread configurations (30 records, 0 mismatches), plus six $l = 13$ sample records. The recorded PS-2 program hashes reproduce on every rung (`programs_equal_ps2.txt`). Lane utilisation at $l = 11$: 0.913, 0.780, 0.653 (64, 256, 512 lanes).

3. Missing or not done

The thread stage; $l = 13$ rounds 4 and 5; the 4-thread isolated run (the tool refuses to reserve all four vCPUs, see the host B report); `bs512` under `callgrind` (valgrind 3.22 does not execute AVX-512). Wall figures are exploratory in the repository's isolation vocabulary. Nothing end to end; nothing for ECC2K-130. Host B rows are in a separate report and are not pooled with these.